

FUNCTIONAL REQUIREMENTS

R1: Question Bank and Adversarial Repository Management (UC1)

R1.1: Question Creation

R1.1.1: Allow a recruiter to create a new source question.

R1.1.2: Allow a recruiter to categorise a question by category, difficulty level, question type, and tags during creation.

R1.1.3: Allow a recruiter to store all created source questions in the question bank.

R1.2: Question Editing

R1.2.1: Allow a recruiter to edit the title, content, category, difficulty level, question type, and tags of an existing source question.

R1.3: Question Deletion

R1.3.1: Allow a recruiter to delete a source question from the question bank.

R1.4: Question Search and Filtering

R1.4.1: Allow a recruiter to search and filter the question bank by keyword, category, difficulty level, question type and tags.

R1.5: Repository Management

R1.5.1: Maintain a source question repository using the Question Bank table to store all original questions.

R1.5.2: Maintain a separate adversarial question repository using the Adversarial Questions table to store all generated adversarial versions.

R1.5.3: Maintain a reference between the source question and adversarial question.

R2: Adversarial Question Generation (UC2)

R2.1: Generation Triggering

R2.1.1: Allow a recruiter to trigger adversarial question generation for source questions.

R2.2: Question Generation Process

R2.2.1: Retrieve the selected source question content from the question bank table when adversarial generation is triggered.

R2.2.2 Construct a structured adversarial prompt template and submit it along with the source question content to the Google Gemini API.

R2.2.3 Create an adversarial question using Google Gemini which will embed adversarial characteristics into the question.

R2.2.4 Display the adversarial version of a question to the candidate during assessment delivery.

R2.3: Question Generation Preview and Review

R2.3.1: Allow the recruiter to regenerate an adversarial version if the

initial result is unsatisfactory.

R3: Adversarial Question Verification (UC3)

R3.1: Verification Triggering

R3.1.1: Allow a recruiter to trigger the adversarial question verification process after an adversarial question has been generated.

R3.1.2: Allow a recruiter to verify an adversarial question before the question is deployed to an assessment.

R3.2: Data Retrieval

R3.2.1: Retrieve the corresponding source question for the generated adversarial question for verification.

R3.2.2 Retrieve the stored correct answer for the generated adversarial question.

R3.3: Reference Answer Generation

R3.3.1: Retrieve a reference answer representing what the Google Gemini would return for that generated adversarial question.

R3.4: Manual verification by Recruiter

R3.4.1: Display a side by side screen of the source question and the generated adversarial question to ensure the same core skill is still being tested.

R3.4.2: Display a side by side screen of the stored correct answer and the Google Gemini reference answer for the adversarial question for validity.

R3.4.3: Allow a recruiter to either approve or reject the adversarial question.

R3.5: Verification Outcome

R3.5.1: Approved adversarial questions can be added to candidate assessments.

R3.5.2: Rejected adversarial questions cannot be added to candidate assessments.

R4: Code Execution and Improved Assessment Experience (UC4)

R4.1: Assessment Visibility

R4.1.1: Allow a candidate to view all assessments that have been assigned to them on the candidate dashboard

R4.2: Assessment Resumption

R4.2.1: Allow a candidate to resume an incomplete assessment from the point at which they left off.

R4.3: Question Answering

R4.3.1: Allow a candidate to answer coding questions through an integrated code editor supporting Python.

R4.4: Code execution

R4.4.1: Upon submission of a coding question, the system should send the candidate's Python code to Piston CE for execution.

R4.4.2 Retrieve and execute appropriate test cases against the candidate's submission.

R4.5: Scoring Processing and Storage

R4.5.1: Calculate and update scoring information for individual responses and overall assessments.

R4.6: Assessment Submission

R4.6.1: Allow a candidate to submit their completed assessment.

R5: Custom Assessment Creation (UC5)

R5.1: Assessment Creation

R5.1.1: Allow a recruiter to create a new assessment by providing a title, description and duration.

R5.2: Adversarial Question Selection

R5.2.1: Allow a recruiter to select adversarial questions to add to the assessment.

R5.3: Assessment Publishing

R5.3.1: Allow the recruiter to publish and finalise assessment creation.

R5.4: Assessment Assignment

R5.4.1: Allow a recruiter to assign a completed assessment to one or more candidates.

R5.4.2: Generate a unique access token per candidate assessment assignment.